ModelIt!

Teacher's Guide

How Do Self-Driving Cars See?

Modeling LiDAR, Computer Vision, and Autonomous Decision-Making

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-PS4-1, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS4-1, HS-ETS1-2

Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models of sensor fusion and autonomous perception to predict how changes in sensor configuration, environmental conditions, and processing parameters affect driving safety.
Disciplinary Core Idea	**PS4.A Wave Properties / ETS1.B Developing Possible Solutions** — Students model how electromagnetic waves (laser light, radio waves) interact with matter to create sensor measurements, and engineer solutions to the perception and decision-making challenges of autonomous driving.
Crosscutting Concept	**Systems and System Models** — Students analyze the autonomous vehicle as an integrated system where sensors, processors, and actuators must work together with precise timing, and where failure in any subsystem cascades through the entire driving system.

CAST Testing Alignment

MULTIPLE CHOICE:

An autonomous vehicle traveling at 65 mph has a total processing latency of 300 milliseconds from sensor input to actuator response. If a pedestrian steps into the road 150 feet ahead, and the vehicle requires 120 feet to stop at this speed, does the vehicle have sufficient distance to stop safely? Show your calculation including the distance traveled during the processing delay.

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why autonomous vehicles use multiple sensor types rather than relying solely on the highest-resolution sensor. Address how environmental conditions affect different sensors differently and how sensor fusion provides reliability that no single sensor can achieve alone.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every time you ride in a car, your brain performs an extraordinary feat of real-time perception: processing visual information from two eyes, interpreting depth, motion, color, and context, predicting the behavior of other road users, and making continuous steering and speed decisions — all while holding a conversation or listening to music. Autonomous vehicles attempt to replicate this capability using an array of electronic sensors and artificial intelligence algorithms. The engineering challenge is immense: the system must work perfectly in conditions ranging from bright sunshine to heavy rain, from empty highways to crowded school zones, and must handle situations that no programmer anticipated.

LiDAR: Seeing in 3D with Light

LiDAR sensors emit thousands of short laser pulses per second — typically 905 nanometer or 1550 nanometer infrared light — and measure the precise time each pulse takes to return after reflecting off objects. Because light travels at a known constant speed (approximately 300 million meters per second), the round-trip time reveals the exact distance to each reflection point. A modern automotive LiDAR unit generates 300,000 to 2 million points per second, creating a detailed 3D point cloud of everything within 200 meters. This point cloud provides precise geometric information — the shape, size, and distance of every object — but contains no color or texture information. LiDAR works equally well in daylight and darkness because it generates its own illumination, but performance degrades in rain, fog, and snow because water droplets scatter the laser beams, creating false returns and reducing range.

Computer Vision: Understanding What the Sensors See

Cameras provide rich color and texture data that complement LiDAR's geometric precision. Convolutional neural networks trained on millions of labeled images can classify objects — distinguishing a child from a fire hydrant, a stop sign from a billboard, a pedestrian crossing from a shadow on the road. Modern autonomous vehicles use 8-12 cameras covering a full 360-degree view, often with both wide-angle and telephoto lenses. The neural networks process each camera frame (typically 30-60 frames per second) through dozens of computational layers that detect edges, textures, shapes, and contextual relationships. However, cameras share human vision's weaknesses: they struggle with glare, darkness, and weather that reduces visibility. They also inherit biases from their training data — a classifier trained primarily on images from sunny California may fail to recognize pedestrians in winter clothing or objects partially obscured by snow.

Sensor Fusion: Building a Unified World Model

No single sensor provides complete information for safe driving. LiDAR provides precise 3D geometry but no color. Cameras provide rich visual detail but imprecise depth. Radar provides velocity measurements and works through weather but has low spatial resolution. Sensor fusion algorithms combine these complementary data streams into a single unified environmental model. The fusion process uses probabilistic methods — typically Kalman filters or particle filters — to weight each sensor's contribution based on its reliability in current conditions. When a LiDAR point cloud shows an object at 50 meters and a camera classifies it as a pedestrian, the fused model represents a pedestrian at 50 meters with high confidence. When sensors disagree — perhaps the camera sees nothing but LiDAR detects something — the system must resolve the conflict, usually by increasing uncertainty and triggering cautious behavior. The computational cost of real-time sensor fusion is enormous, requiring specialized processors that consume hundreds of watts of power.

The Decision-Making Challenge

Perception — knowing what is around the car — is only half the problem. The vehicle must also decide what to do. Path-planning algorithms calculate the optimal trajectory considering road geometry, traffic rules, other vehicles' predicted movements, and safety constraints. In normal driving, this is relatively straightforward: follow the lane, maintain safe distance, obey signals. The challenge explodes in complexity at intersections, merges, and construction zones where multiple agents interact with conflicting intentions. And then there are edge cases: a ball rolling into the street (a child may follow), an emergency vehicle approaching from behind, a road surface that looks like a lane marking, or a completely novel situation the system has never encountered. Human drivers handle these with general intelligence, social awareness, and common sense.

Current autonomous systems handle them with statistical pattern matching that can fail catastrophically when reality deviates from training data. This gap between statistical perception and genuine understanding is the central unsolved challenge of autonomous driving.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven key components of an autonomous vehicle perception system: LiDAR resolution, camera detection accuracy, sensor fusion confidence, processing latency, environmental degradation, decision algorithm reliability, and safety margin.

E **Establish Relationships**

Students map relationships showing that LiDAR resolution and environmental degradation are the primary drivers of sensor fusion confidence, which combined with processing latency determines the safety margin available for decision-making at any given speed.

V **Visualize & Model**

Students build a computational model showing the seven-component autonomous perception system with sensor data flowing through fusion into decision-making.

E **Evaluate Outcomes**

Students run clear highway, heavy rain urban, and novel object emergency scenarios to discover the conditions under which the autonomous system maintains safe operation versus requiring human intervention.

R **Revise & Extend**

Students add V2X communication, prediction horizon, or ethical decision framework to model more realistic autonomous driving complexity and edge-case handling.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, cars are driving themselves on public roads. They use lasers, cameras, and radar to see the world, and artificial intelligence to decide what to do. They are already safer than average human drivers in some conditions — and catastrophically confused in others. How do they actually work?"*

Watch For

Open with a video clip or image of an autonomous vehicle in operation. Ask: who has seen a self-driving car? Who would ride in one? Who would NOT ride in one? Catalog the trust concerns.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the perception and decision system of an autonomous vehicle. You will discover why these systems work brilliantly most of the time and fail dangerously in edge cases — and why that gap might be the hardest engineering problem of the century."*

Watch For

Have students read objectives. Pre-teach 'point cloud' with a visual showing raw LiDAR data. Pre-teach 'decision latency' by calculating distance traveled at highway speed during a 300ms delay.

Slide 3: Big Question

3 min

Say: *"Your brain does this effortlessly — you glance at a street and instantly understand the scene. But your brain has 86 billion neurons refined by 500 million years of evolution. An autonomous vehicle has sensors, processors, and algorithms designed in the last twenty years. Let us see how close they get."*

Watch For

Quick-write: Students estimate how many data points per second a self-driving car must process. Reveal the real number (millions) and discuss the computational challenge.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where multiple sensors must agree on what is real, where milliseconds determine safety, and where the hardest problems are the situations nobody predicted."*

Watch For

Preview LEVER steps. Emphasize that this model connects physics (how sensors work) to engineering (how data is processed) to ethics (how decisions are made).

Slide 5: Activity 1: Components

8 min

Say: *"Seven components. Two you control: LiDAR resolution and environmental conditions. Five that emerge from physics, algorithms, and engineering constraints. The key insight: safety margin is what is left over after every other component takes its toll."*

Watch For

Guide through components. Demonstrate LiDAR point cloud density at different resolutions. Show how environmental degradation affects each sensor type differently.

Slide 6: Activity 2: Connections

8 min

Say: *"Every sensor feeds into fusion, fusion feeds into decisions, and decisions must happen faster than the car covers dangerous distance. The bottleneck is not any single sensor — it is the entire pipeline from photon to steering wheel."*

Watch For

Help students map the data flow pipeline. Calculate the distance traveled during processing latency at different speeds. Highlight the sensor disagreement problem.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios that test the system from comfortable to catastrophic. Watch what happens to your safety margin as conditions get harder."*

Watch For

Students predict safety margins before each scenario. The novel object scenario is the key insight — the system cannot safely handle what it cannot classify. Discuss what the vehicle should do when confused.

Slide 8: Discoveries

5 min

Say: *"Self-driving cars are engineering marvels that work brilliantly in the 95% of driving that is routine. The challenge — and the reason full autonomy remains elusive — is the 5% of situations that require human-level understanding of context, intention, and common sense."*

Watch For

Lead discussion on the gap between statistical pattern matching and genuine understanding. When is good enough actually good enough for public safety? How do we quantify acceptable risk?

Slide 9: STEM Challenge

12 min

Say: *"A real company needs your sensor design for their autonomous delivery vehicles in YOUR city. You know the weather, the roads, the traffic patterns. Design the system."*

Watch For

Groups design sensor suites for their local conditions. Must specify sensor types and positions, processing requirements, safety thresholds, and failure protocols. Present with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** LiDAR Resolution, Environmental Degradation
- **INTERNAL:** Camera Detection Accuracy, Sensor Fusion Confidence, Processing Latency, Decision Algorithm Reliability, Safety Margin

Relationships

- **LiDAR Resolution to Sensor Fusion Confidence:** POSITIVE (+) — Higher LiDAR resolution provides more detailed 3D point clouds with more data points per object, giving the fusion algorithm more geometric information to work with and increasing the confidence of the unified environmental model.
- **Environmental Degradation to Camera Detection Accuracy:** NEGATIVE (-) — Worsening environmental conditions — rain, fog, darkness, glare — directly reduce the quality of camera images and the accuracy of neural network object classification, because the visual features the network relies on become obscured or distorted.
- **Processing Latency to Safety Margin:** NEGATIVE (-) — Longer processing latency means the vehicle travels farther between sensing and acting, directly consuming safety margin. At 65 mph, each additional 100 milliseconds of latency reduces the available stopping distance by approximately 10 feet.
- **Sensor Fusion Confidence to Decision Algorithm Reliability:** POSITIVE (+) — Higher fusion confidence means the decision algorithm receives a more accurate environmental model, enabling more reliable path planning. When fusion confidence is low, the decision algorithm must make choices based on uncertain information, increasing the probability of errors.

Key Discoveries

- Sensor fusion is not simply averaging data from different sensors — it is a probabilistic calculation where each sensor's contribution is weighted by its reliability in current conditions, and the system's overall confidence can be lower than any individual sensor's confidence when sensors disagree
- Processing latency creates a fundamental safety limit: at highway speeds, the vehicle is always responding to a world that existed hundreds of milliseconds ago, meaning safety margins must account for everything that could change during the processing delay
- Environmental conditions do not degrade all sensors equally — rain primarily affects LiDAR and cameras but barely affects radar, while darkness eliminates camera utility but does not affect LiDAR or radar — which is why redundant multi-sensor systems are essential for all-weather autonomy
- The hardest challenge for autonomous vehicles is not normal driving but edge cases — the unusual, rare, and novel situations that were not well-represented in training data, where human drivers use general intelligence and common sense that current AI systems lack

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Self-driving cars see the world the same way humans do"	Autonomous vehicles and humans use fundamentally different perception mechanisms. Humans process visual information through biological neural networks refined by millions of years of evolution, creating a rich understanding of context, intention, and social cues. Autonomous vehicles process discrete sensor measurements through mathematical algorithms that detect statistical patterns without understanding meaning. A human sees a person looking at their phone while walking toward a crosswalk and predicts they might step into traffic without looking. An autonomous system sees a cluster of LiDAR points classified as a pedestrian and predicts movement based on trajectory statistics. The model shows how this gap manifests in edge cases where contextual understanding matters.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More sensors always make the car safer"	Additional sensors provide more data but also increase processing latency, system complexity, and potential points of failure. Each new sensor must be integrated into the fusion algorithm, calibrated against all other sensors, and its data processed in real time. Beyond a certain point, the added processing delay from handling more data can actually reduce safety margin more than the improved perception adds. The model demonstrates the optimal sensor count where the trade-off between perception improvement and processing latency reaches its best balance.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Autonomous vehicles just need more training data to handle edge cases"	More training data improves performance on situations similar to what has been seen before, but it cannot prepare the system for truly novel situations. The real world has an effectively infinite number of possible scenarios, and no dataset — however large — can include them all. A car trained on millions of hours of driving data in temperate climates may still fail when encountering its first dust storm, its first parade, or its first earthquake. The fundamental limitation is not data quantity but the gap between pattern recognition and causal reasoning. The model shows how classification confidence drops to near-random levels for objects and scenarios outside the training distribution.